

ESResSo++: A Fast and Extensible Molecular Simulation Package for Coarse-Grained Models

Zhen-Hao Xu⁴, James Vance⁴, Nikita Tretyakov⁴, Sebastian Eibl², Pavel Kus², Jakub Krajniak⁶, Tristan Bereau⁵, Horacio V. Guzman⁷, Bin Song¹, Markus Rampp², Torsten Stuehn¹, and Christoph Junghans³

¹ Max Planck Institute for Polymer Research, Mainz, Germany ² Max Planck Computing and Data Facility, Garching, Germany ³ Los Alamos National Laboratory, Los Alamos, USA ⁴ Johannes Gutenberg University of Mainz, Mainz, Germany ⁵ Heidelberg University, Heidelberg, Germany ⁶ Independent researcher, Poznań, Poland ⁷ Institut de Ciència de Materials, Barcelona, Spain

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Editor: [Patrick Diehl](#)

Reviewers:

- [@tingyu66](#)
- [@jngrad](#)

Submitted: 07 March 2026

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

ESResSo++ is an open-source software package for molecular dynamics (MD) simulations with a particular emphasis on coarse-grained (CG) models of soft matter systems (Praprotnik et al., 2008). Written in C++ with a flexible Python interface, it is designed for high-performance computing (HPC) environments and supports massively parallel simulations through MPI. The package enables simulations of polymers, membranes, colloids and complex fluids with a wide range of interaction models and advanced algorithms.

ESResSo++ builds upon the experience of its predecessor ESResSo, but provides a cleaner, more modular codebase and enhanced extensibility. It is actively developed by an international community of researchers across multiple institutions and disciplines.

Statement of need

Molecular dynamics simulations are essential tools for exploring the behavior of soft matter systems at mesoscopic scales. General-purpose MD codes such as GROMACS (Abraham et al., 2015) and LAMMPS (Thompson et al., 2022) are primarily optimized for all-atom simulations and may require significant customization for coarse-grained models that need non-standard interactions or specialized multiscale algorithms. ESResSo++ addresses this gap by providing:

- A modular and extensible design, enabling researchers to easily implement new interaction potentials and integrators.
- Efficient parallelization for large-scale simulations of complex systems.
- A rich library of coarse-grained interaction models and algorithms tailored to soft matter.
- A Python-based scripting interface for ease of use, reproducibility, and coupling with external analysis tools.
- Multi-scale simulation techniques such as AdResS and Lees-Edwards.

State of the field

Several established MD packages serve the soft matter and coarse-grained simulation communities. LAMMPS (Thompson et al., 2022) is a general-purpose, highly scalable code with broad force field support and a flexible plugin architecture; however, its input scripting language is less suited to complex CG workflows, and adaptive resolution is not a core feature. GROMACS (Abraham et al., 2015) provides exceptional performance for standard force

fields and supports the Martini CG model ecosystem, but adding truly novel CG interactions requires modifying the C++ source, and its file-based workflow is less interactive than a Python API. **HOOMD-blue** (Anderson et al., 2020) is the most directly comparable package, offering a Python-native API and GPU-accelerated CG simulations for soft matter; however, it lacks support for adaptive resolution methods. **OpenMM** (Eastman et al., 2017) excels at GPU-accelerated simulations and allows custom force definitions via algebraic expressions, but targets primarily biomolecular systems and lacks MPI-based multi-node parallelization. The sibling project **ESPResSo** (Weik et al., 2019) shares common roots but focuses on charged systems, hydrodynamic coupling (lattice Boltzmann), and electrokinetics rather than multiscale resolution bridging.

ESPREsSo++ occupies a distinct niche as a package purpose-built for coarse-grained and multiscale soft matter simulations. Its key differentiator is first-class support for adaptive resolution simulation (AdResS), which allows seamless coupling of atomistic and coarse-grained representations within a single simulation. ESPREsSo++ is currently the only actively maintained MD package that provides AdResS as a core feature; none of the other packages listed above offer this capability. It is worth noting that these capabilities were also contributed to existing general-purpose codes (Fritsch et al., 2012; Junghans & Pobleto, 2010; Nagarajan et al., 2013), but got removed again later, hence a dedicated package allows tighter integration of multiscale algorithms with the domain decomposition, neighbor list construction, and force computation pipeline — design choices that would be difficult to retrofit into architectures optimized for all-atom throughput.

60 Software design

ESPResSo++ uses a layered C++/Python architecture. The performance-critical computational kernel (force calculations, integration, domain decomposition, and communication) is implemented in C++17 and exposed to Python through Boost.Python bindings. Each C++ class provides a static registerPython() method, and a central registration dispatcher ensures that the full object hierarchy is available as the espressopp Python module. This design lets users assemble, configure, and control simulations entirely from Python while retaining the performance of compiled C++ for the inner loops.

68 **Modularity through templates and extensions.**

Interactions are implemented using C++ class templates parameterized by a potential type (e.g., `VerletListInteractionTemplate<_Potential>`), so that adding a new pairwise potential requires only implementing the `computeEnergy()` and `computeForce()` methods of a `Potential` subclass. The integrator follows a similar pattern: an `MDIntegrator` base class exposes an `addExtension()` mechanism through which thermostats, barostats, constraints, and adaptive resolution layers are attached via `Boost.Signals2` callbacks. This signal-based coupling keeps extensions decoupled from the integration loop and from each other.

76 **Parallelization.**

ESPResSo++ employs spatial domain decomposition with MPI. The simulation box is partitioned across MPI ranks using a `NodeGrid`, and each rank further subdivides its domain into linked cells via a `CellGrid`. Ghost particle exchange handles communication of boundary data. A non-blocking variant (`DomainDecompositionNonBlocking`) overlaps communication with computation, and the heterogeneous spatial domain decomposition algorithm (HeSpaDDA) (Guzman et al., 2017) provides load balancing for spatially inhomogeneous systems.

83 Performance optimizations.

Since version 2.0 (Guzman et al., 2019), ESPResSo++ has been modernized with SIMD
vectorization through a structure-of-arrays (SOA) particle data layout (ParticleArray) with
64-byte alignment, yielding an overall three-times speedup for short-range non-bonded force
calculations (Vance et al., 2023). An improved cell decomposition scheme (Yao et al., 2004)
allows sub-decomposition into cells with a length of half or a third of the cutoff radius, reducing

89 the number of unnecessary distance calculations.

90 **Testing and documentation.**

91 The code is tested through a combination of Boost.Test (C++) and Python unittest test
92 suites, executed via CMake/CTest and continuous integration on GitHub Actions. User
93 documentation is built with Sphinx; developer API documentation with Doxygen.

94 **Research impact statement**

95 ESPResSo++ has enabled research across a broad range of soft matter topics over more than
96 a decade of active development. It has been used in numerous peer-reviewed studies, including
97 investigations of:

- 98 ■ Polymer rheology and entanglement effects ([Grommes et al., 2021, 2022, 2024, 2025](#);
99 [Grommes & Reith, 2020](#); [Hsu & Kremer, 2020, 2023, 2024](#); [Lee & Paul, 2020](#); [Ohkuma](#)
100 [et al., 2023](#); [Singh et al., 2020](#); [Tubiana et al., 2021](#); [Zhao, Singh, et al., 2020](#))
- 101 ■ Lipid membranes, protein and vesicle dynamics ([Bause & Bereau, 2021](#); [Papež et al.,](#)
102 [2023](#); [Zhao, Cortes-Huerto, et al., 2020](#))
- 103 ■ Adaptive resolution simulations ([Fiorentini et al., 2020](#); [Thaler et al., 2020](#))
- 104 ■ Ionic liquids under shear flow ([Gholami et al., 2025](#); [Zhang et al., 2021](#))
- 105 ■ Coarse-grained conformational surface hopping ([Rudzinski & Bereau, 2020](#))

106 The project is developed across multiple institutions — Max Planck Institute for Polymer
107 Research, Max Planck Computing and Data Facility, Los Alamos National Laboratory, Johannes
108 Gutenberg University of Mainz, and Heidelberg University — with contributions from both
109 current and former developers documented in the AUTHORS file. The codebase has a public
110 development history spanning more than ten years on GitHub, with continuous integration,
111 code coverage tracking, and versioned releases.

112 **Functionality**

113 Key features of ESPResSo++ include:

- 114 ■ **Inter-particle interactions:** Lennard-Jones, Coulomb, soft repulsive, bonded interactions,
115 tabulated potentials, and more.
- 116 ■ **Algorithms:** Molecular dynamics, Langevin dynamics, dissipative particle dynamics
117 (DPD), Brownian dynamics, adaptive resolution simulations (AdResS), Monte Carlo
118 sampling.
- 119 ■ **Electrostatics:** Particle–particle particle–mesh (P3M), Ewald summation, and other
120 long-range methods.
- 121 ■ **Parallelization:** Domain decomposition using MPI, optimized for massively parallel
122 architectures.
- 123 ■ **Python interface:** Full simulation control and analysis scripting in Python.
- 124 ■ **Extensibility:** Modular design allows easy addition of new force fields, integrators, or
125 analysis tools.

126 **New Features since last release**

127 Since the last major release of ESPResSo++ v2.0 in 2018 ([Guzman et al., 2019](#)) a number of
128 new functionalities and features have been added, including:

- 129 ■ **SIMD vectorization and related optimizations:** enhance compute performance on modern
130 CPUs ([Vance et al., 2023](#))
- 131 ■ **Cell decomposition:** allow sub-decomposition into cells with a length of half or a third
132 of the cutoff for direct force calculations ([Yao et al., 2004](#))

- **HeSpaDDA**: heterogeneous spatial domain decomposition algorithm (HeSpaDDA) for larger scale simulations (Guzman et al., 2017)
- **new potentials and simulation methods**: AngularCosineSquared, TabulatedSubEnsAngular, surface hopping MD, Lees-Edwards boundary conditions
- **Checkpoint the state of the random number generator (RNG)**: allow restarting from checkpointed state of RNG
- **I/O**: support for parallel writing and reading of H5MD checkpoints
- **Python 3 compatibility**

Example usage

A minimal Python script to run a simple (repulsive only) Lennard-Jones type particle simulation in ESPReso++ looks like:

```
import espressopp
from mpi4py import MPI

# simulation system parameters
num_particles = 10000      # total number of particles in the system
box            = (20,20,20) # size of the simulationbox (all length are in sigma)
rc             = 1.12246   # cut off for the short range non bonded potential
skin          = 0.3        # skin used for verlet neighbor list
dt            = 0.005      # time step for 1 md step
epsilon       = 1.0        # energy unit
sigma         = 1.0        # length unit
temperature   = 1.0        # temperature of the simulation
LJcaprad      = 0.8        # initial capping radius for LJ interaction
                                # for random configurations

# system setup
system        = espressopp.System( )
system.rng     = espressopp.esutil.RNG( )
system.bc     = espressopp.bc.OrthorhombicBC(system.rng, box)
system.skin   = skin

# define underlying storage system for parallelisation
nodeGrid      = espressopp.tools.decomp.nodeGrid(MPI.COMM_WORLD.size,box,rc,skin)
cellGrid      = espressopp.tools.decomp.cellGrid(box, nodeGrid, rc, skin)
system.storage = espressopp.storage.DomainDecomposition(system, nodeGrid, cellGrid)

# interaction setup, here short range non-bonded Lennard Jones potential
interaction    = espressopp.interaction.VerletListLennardJonesCapped(
                    espressopp.VerletList(system, cutoff=rc))
interaction.setPotential(type1=0, type2=0,
                        potential=espressopp.interaction.LennardJonesCapped(
                            epsilon, sigma, shift='auto', caprad=LJcaprad))
system.addInteraction(interaction)

# integrator setup
integrator     = espressopp.integrator.VelocityVerlet(system)
integrator.dt  = dt

# thermostat setup
thermostat     = espressopp.integrator.LangevinThermostat(system)
thermostat.gamma = 1.0
```

```

thermostat.temperature = temperature
integrator.addExtension(thermostat)

# create random particle setup in the simulation box
props = ['id', 'type', 'mass', 'pos', 'v']
new_particles = []
pid = 1
while pid <= num_particles:
    type = 0
    mass = 1.0
    pos = system.bc.getRandomPos()
    vel = espressopp.Real3D(0.0, 0.0, 0.0)
    part = [pid, type, mass, pos, vel]
    new_particles.append(part)
    if pid % 1000 == 0:
        system.storage.addParticles(new_particles, *props)
        system.storage.decompose()
        new_particles = []
    pid += 1
system.storage.addParticles(new_particles, *props)

integrator.dt = 0.0001 # very small timestep for initial warmup
for n in range(20):
    # warmup finished, switch to uncapped Lennard Jones potential and increase timestep
    if n == 10:
        interaction = espressopp.interaction.VerletListLennardJones(
            espressopp.VerletList(system, cutoff=rc))
        interaction.setPotential(type1=0, type2=0,
                                potential=espressopp.interaction.LennardJones(
                                    epsilon, sigma, rc, shift='auto'))
        system.removeInteraction(0)
        system.addInteraction(interaction)
        integrator.dt = 0.005
    integrator.run(10000)
    Etot = system.getInteraction(0).computeEnergy()
    print("md time = {:.4f}, total energy: {:.10.2f}".format(integrator.dt*n*10000, Etot))

# write PDB file of (quasi) equilibrated LJ system.
# At this temperature it is more or less crystallized.
espressopp.tools.pdbwrite("simplelj.pdb", system, molsize=num_particles)

```

AI usage disclosure

No generative AI tools were used in the creation of the ESPResSo++ software or its documentation. During the preparation of this manuscript, AI-assisted tools were used for copyediting and formatting. All content was reviewed and verified by the authors.

Acknowledgements

We thank the ESPResSo++ developer community and all contributors listed in the AUTHORS file. ESPResSo++ has been supported by the Transregio TRR146 of the German Research Foundation. The ESPResSo++ project is supported by the U.S. Department of Energy through Los Alamos National Laboratory (LANL). Los Alamos National Laboratory is operated by

- 153 Triad National Security, LLC, for the National Nuclear Security Administration of the U.S.
 154 Department of Energy (contract no. 89233218CNA000001). This paper has been assigned a
 155 Los Alamos Unlimited Release number of LA-UR-26-21700.
- 156 Abraham, M. J., Murtola, T., Schulz, R., Páll, S., Smith, J. C., Hess, B., & Lindahl, E. (2015).
 157 GROMACS: High performance molecular simulations through multi-level parallelism from
 158 laptops to supercomputers. *SoftwareX*, 1–2, 19–25. [https://doi.org/10.1016/j.softx.2015.](https://doi.org/10.1016/j.softx.2015.06.001)
 159 [06.001](https://doi.org/10.1016/j.softx.2015.06.001)
- 160 Anderson, J. A., Glaser, J., & Glotzer, S. C. (2020). HOOMD-blue: A python package for high-
 161 performance molecular dynamics and hard particle monte carlo simulations. *Computational*
 162 *Materials Science*, 173, 109363. <https://doi.org/10.1016/j.commatsci.2019.109363>
- 163 Bause, M., & Bereau, T. (2021). Reweighting non-equilibrium steady-state dynamics along
 164 collective variables. *The Journal of Chemical Physics*, 154(13). [https://doi.org/10.1063/5.](https://doi.org/10.1063/5.0042972)
 165 [0042972](https://doi.org/10.1063/5.0042972)
- 166 Eastman, P., Swails, J., Chodera, J. D., McGibbon, R. T., Zhao, Y., Beauchamp, K. A.,
 167 Wang, L.-P., Simmonett, A. C., Harrigan, M. P., Stern, C. D., Wiewiora, R. P., Brooks,
 168 B. R., & Pande, V. S. (2017). OpenMM 7: Rapid development of high performance
 169 algorithms for molecular dynamics. *PLOS Computational Biology*, 13(7), e1005659.
 170 <https://doi.org/10.1371/journal.pcbi.1005659>
- 171 Fiorentini, R., Kremer, K., & Potestio, R. (2020). Ligand-protein interactions in lysozyme
 172 investigated through a dual-resolution model. *Proteins: Structure, Function, and*
 173 *Bioinformatics*, 88(10), 1351–1360. <https://doi.org/10.1002/prot.25954>
- 174 Fritsch, S., Junghans, C., & Kremer, K. (2012). Structure formation of toluene around C60:
 175 Implementation of the adaptive resolution scheme (AdResS) into GROMACS. *Journal of*
 176 *Chemical Theory and Computation*, 8(2), 398–403. <https://doi.org/10.1021/ct200706f>
- 177 Gholami, A., Kloth, S., Xu, Z.-H., Kremer, K., Vogel, M., Stuehn, T., & Rudzinski, J. F.
 178 (2025). Structure and dynamics of ionic liquids under shear flow. *The Journal of Chemical*
 179 *Physics*, 163(7). <https://doi.org/10.1063/5.0279946>
- 180 Grommes, D., Bruch, O., Imhof, W., & Reith, D. (2025). Coarse-grained molecular dynamics
 181 study of the melting dynamics in long alkanes. *Polymers*, 17(18). [https://doi.org/10.](https://doi.org/10.3390/polym17182500)
 182 [3390/polym17182500](https://doi.org/10.3390/polym17182500)
- 183 Grommes, D., Bruch, O., & Reith, D. (2024). Mimicking polymer processing conditions on the
 184 meso-scale: Relaxation and crystallization in polyethylene systems after uni- and biaxial
 185 stretching. *Molecules*, 29(14), 3391. <https://doi.org/10.3390/molecules29143391>
- 186 Grommes, D., & Reith, D. (2020). Determination of relevant mechanical properties for the
 187 production process of polyethylene by using mesoscale molecular simulation techniques.
 188 *Soft Materials*, 18(2–3), 242–261. <https://doi.org/10.1080/1539445x.2020.1722692>
- 189 Grommes, D., Schenk, M. R., Bruch, O., & Reith, D. (2021). Investigation of crystallization
 190 and relaxation effects in coarse-grained polyethylene systems after uniaxial stretching.
 191 *Polymers*, 13(24), 4466. <https://doi.org/10.3390/polym13244466>
- 192 Grommes, D., Schenk, M. R., Bruch, O., & Reith, D. (2022). Initial crystallization effects
 193 in coarse-grained polyethylene systems after uni- and biaxial stretching in blow-molding
 194 cooling scenarios. *Polymers*, 14(23), 5144. <https://doi.org/10.3390/polym14235144>
- 195 Guzman, H. V., Junghans, C., Kremer, K., & Stuehn, T. (2017). Scalable and fast
 196 heterogeneous molecular simulation with predictive parallelization schemes. *Physical*
 197 *Review E*, 96, 053311. <https://doi.org/10.1103/PhysRevE.96.053311>
- 198 Guzman, H. V., Tretyakov, N., Kobayashi, H., Fogarty, A. C., Kreis, K., Krajniak, J., Junghans,
 199 C., Kremer, K., & Stuehn, T. (2019). ESPResSo++ 2.0: Advanced methods for multiscale
 200 molecular simulation. *Computer Physics Communications*, 238, 66–76. <https://doi.org/10.1016/j.cpc.2019.06.031>

- 1016/j.cpc.2018.12.017
- Hsu, H.-P., & Kremer, K. (2020). Efficient equilibration of confined and free-standing films of highly entangled polymer melts. *The Journal of Chemical Physics*, 153(14). <https://doi.org/10.1063/5.0022781>
- Hsu, H.-P., & Kremer, K. (2023). Glass transition temperature of (ultra-)thin polymer films. *The Journal of Chemical Physics*, 159(7). <https://doi.org/10.1063/5.0165902>
- Hsu, H.-P., & Kremer, K. (2024). Entanglement-stabilized nanoporous polymer films made by mechanical deformation. *Macromolecules*, 57(6), 2998–3012. <https://doi.org/10.1021/acs.macromol.4c00187>
- Junghans, C., & Poblete, S. (2010). A reference implementation of the adaptive resolution scheme in ESPResSo. *Computer Physics Communications*, 181(8), 1449–1454. <https://doi.org/10.1016/j.cpc.2010.04.013>
- Lee, E., & Paul, W. (2020). Additional entanglement effect imposed by small sized ring aggregates in supramolecular polymer melts: Molecular dynamics simulation study. *Macromolecules*, 53(5), 1674–1684. <https://doi.org/10.1021/acs.macromol.9b02209>
- Nagarajan, A., Junghans, C., & Matysiak, S. (2013). Multiscale simulation of liquid water using a four-to-one mapping for coarse-graining. *Journal of Chemical Theory and Computation*, 9(11), 5168–5175. <https://doi.org/10.1021/ct400566j>
- Ohkuma, T., Hagita, K., Murashima, T., & Deguchi, T. (2023). Miscibility and exchange chemical potential of ring polymers in symmetric ring–ring blends. *Soft Matter*, 19(21), 3818–3827. <https://doi.org/10.1039/d3sm00108c>
- Papež, P., Merzel, F., & Praprotnik, M. (2023). Rotational dynamics of a protein under shear flow studied by the eckart frame formalism. *The Journal of Physical Chemistry B*, 127(33), 7231–7243. <https://doi.org/10.1021/acs.jpcc.3c02324>
- Praprotnik, M., Junghans, C., Delle Site, L., & Kremer, K. (2008). Simulation approaches to soft matter: Generic statistical properties vs. Chemical details. *Computer Physics Communications*, 179(1–3), 51–60. <https://doi.org/10.1016/j.cpc.2008.01.018>
- Rudzinski, J. F., & Bereau, T. (2020). Coarse-grained conformational surface hopping: Methodology and transferability. *The Journal of Chemical Physics*, 153(21). <https://doi.org/10.1063/5.0031249>
- Singh, M. K., Hu, M., Cang, Y., Hsu, H.-P., Therien-Aubin, H., Koynov, K., Fytas, G., Landfester, K., & Kremer, K. (2020). Glass transition of disentangled and entangled polymer melts: Single-chain-nanoparticles approach. *Macromolecules*, 53(17), 7312–7321. <https://doi.org/10.1021/acs.macromol.0c00550>
- Thaler, S., Praprotnik, M., & Zavadlav, J. (2020). Back-mapping augmented adaptive resolution simulation. *The Journal of Chemical Physics*, 153(16). <https://doi.org/10.1063/5.0025728>
- Thompson, A. P., Aktulga, H. M., Berger, R., Bolintineanu, D. S., Brown, W. M., Crozier, P. S., Veld, P. J. in 't, Kohlmeyer, A., Moore, S. G., Nguyen, T. D., Shan, R., Stevens, M. J., Tranchida, J., Trott, C., & Plimpton, S. J. (2022). LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. *Computer Physics Communications*, 271, 108171. <https://doi.org/10.1016/j.cpc.2021.108171>
- Tubiana, L., Kobayashi, H., Potestio, R., Dünweg, B., Kremer, K., Virnau, P., & Daoulas, K. (2021). Comparing equilibration schemes of high-molecular-weight polymer melts with topological indicators. *Journal of Physics: Condensed Matter*, 33(20), 204003. <https://doi.org/10.1088/1361-648x/abf20c>
- Vance, J., Xu, Z.-H., Tretyakov, N., Stuehn, T., Rampp, M., Eibl, S., Junghans, C., &

- 248 Brinkmann, A. (2023). Code modernization strategies for short-range non-bonded molecular
249 dynamics simulations. *Computer Physics Communications*, 290, 108760. [https://doi.org/](https://doi.org/10.1016/j.cpc.2023.108760)
250 [10.1016/j.cpc.2023.108760](https://doi.org/10.1016/j.cpc.2023.108760)
- 251 Weik, F., Weeber, R., Szuttor, K., Breitsprecher, K., Graaf, J. de, Kuber, M., Kuron, J.,
252 McNamara, S. P., Menzel, A., & Holm, C. (2019). ESPResSo 4.0 – an extensible software
253 package for simulating soft matter systems. *The European Physical Journal Special Topics*,
254 227(14), 1789–1816. <https://doi.org/10.1140/epjst/e2019-800186-9>
- 255 Yao, Z., Wang, J.-S., Liu, G.-R., & Cheng, M. (2004). Improved neighbor list algorithm in
256 molecular simulations using cell decomposition and data sorting method. *Computer Physics*
257 *Communications*, 161(1), 27–35. <https://doi.org/10.1016/j.cpc.2004.04.004>
- 258 Zhang, Z., Krajniak, J., & Ganesan, V. (2021). A multiscale simulation study of influence of
259 morphology on ion transport in block copolymeric ionic liquids. *Macromolecules*, 54(11),
260 4997–5010. <https://doi.org/10.1021/acs.macromol.1c00025>
- 261 Zhao, Y., Cortes-Huerto, R., Kremer, K., & Rudzinski, J. F. (2020). Investigating the
262 conformational ensembles of intrinsically disordered proteins with a simple physics-based
263 model. *The Journal of Physical Chemistry B*, 124(20), 4097–4113. [https://doi.org/10.](https://doi.org/10.1021/acs.jpcc.0c01949)
264 [1021/acs.jpcc.0c01949](https://doi.org/10.1021/acs.jpcc.0c01949)
- 265 Zhao, Y., Singh, M. K., Kremer, K., Cortes-Huerto, R., & Mukherji, D. (2020). Why do
266 elastin-like polypeptides possibly have different solvation behaviors in water–ethanol and
267 water–urea mixtures? *Macromolecules*, 53(6), 2101–2110. [https://doi.org/10.1021/acs.](https://doi.org/10.1021/acs.macromol.9b02123)
268 [macromol.9b02123](https://doi.org/10.1021/acs.macromol.9b02123)

DRAFT